PWM based DC motor speed control system using arduino and L293D motor driver

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

ECA1109 – VLSI design for high performance computing and ai chips

to the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONIC AND COMMUNICATION ENGINEERING

Submitted by

Harini. Veesam (192412566)

Udayini. Angadi (192312158)

Hemalatha (192312157)

Under the Supervision of

Dr. Bala Muragan & Dr. Jacily Jemila



SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

December 2025

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, **[Harini. Veesam, Udayini. Angadi, Hemalatha]** of the ELECTRONIC AND COMMUNICATION ENGINEERING Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled PWM based DC motor speed control system using arduino and L293D motor driver is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: SIMATS ENGINEERING

Date: 11-12-2025

Signature of the Students with Names

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Engineer to Excel

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**PWM BASED DC MOTOR SPEED CONTROL SYSTEM USING ARDUINO AND L293D MOTOR DRIVER**" has been carried out by **[Harini. Veesam, Udayini. Angadi, Hemalatha]** under the supervision of **[**Dr. Bala Muragan & Dr. Jacily Jemila**]** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE	SIGNATURE
Name of the Program	Dr. Bala Muragan
Director	Dr. Jacily Jemila
Program Director	**NAGALAKSHMI MAM**
Saveetha School of	Saveetha School of
Engineering	Engineering
SIMATS	SIMATS

Submitted for the Project work Viva-Voce held on _____

INTERNAL EXAMINER EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, Dr. N.M. Veeraiyan, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, Dr. Deepak Nallaswamy Veeraiyan, and our Vice-Chancellor, Dr. S. Suresh Kumar, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, Dr. Ramya Deepak, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, Dr. B. Ramesh for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, NAGALAKSHMI MAM for his continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guide, Dr. Bala Muragan & Dr.Jacily Jemila for his creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

Signature With Student Name

Harini. Veesam (192412566)

Udayini. Angadi (192312158)

Hemalatha (192312157)

INDEX

S.NO	CONTENT	PAGE NO
1	ABSTRACT	6
2	KEY WORDS	6
3	INTRODUCTION	7
4	METHODOLOGY OVERVIEW	8
5	PROBLEM IDENTIFICATION AND ANALYSIS	10
6	PROBLEM IDENTIFICATION	10
7	SOLUTION DESIGN AND IMPLEMENTATION	11
8	Module 1	13
9	MODULE 2	14
10	Module 3	15
11	Signal OUTPUT	18
12	Result	18
13	FUTURE WORKS	18
14	REFERANCE	20
15	CONCLUSION	20

ABSTRACT

The PWM-based DC motor speed control system using Arduino and the L293D motor driver is designed to provide a reliable, flexible, and energy-efficient solution for controlling motor performance in a wide range of embedded and automation applications. In this project, the Arduino microcontroller generates precise Pulse Width Modulation (PWM) signals whose duty cycle determines the effective voltage applied to the DC motor. By varying the duty cycle, the system achieves smooth, step-less, and highly accurate speed regulation without significant power loss. The L293D motor driver functions as a robust interfacing module, capable of supplying the required current and voltage to the motor while offering built-in protection against back EMF and ensuring safe microcontroller operation.

The system architecture supports both forward and reverse motor rotation, making it suitable for applications where directional control is essential. The use of Arduino enhances programmability, scalability, and real-time responsiveness, allowing easy integration with sensors, displays, and automation modules. This project highlights the advantages of PWM techniques such as reduced heat generation, improved efficiency, and finer control compared to traditional resistive methods. Due to its low cost, simplicity, and high performance, the proposed system is ideal for robotics, conveyor mechanisms, smart appliances, and various mechatronic systems that require reliable motor control. Overall, the project demonstrates an effective combination of embedded programming and motor-driving technology to create a versatile and practical speed control solution.

KEYWORDS:

PWM, DC Motor, Arduino, L293D, Speed Control.

INTRODUCTION

Controlling the speed of a DC motor is an essential requirement in many modern automation, robotics, and industrial applications. Traditional methods of speed control often lead to power loss and poor efficiency, making them unsuitable for systems that require precise and smooth operation. To overcome these limitations, Pulse Width Modulation (PWM) has become one of the most widely used techniques for efficient motor control.

This project, "PWM-Based DC Motor Speed Control System Using Arduino and L293D Motor Driver," focuses on developing a low-cost and reliable solution for adjusting motor speed using PWM signals generated by an Arduino microcontroller. The Arduino varies the duty cycle of the PWM signal, which in turn controls the average voltage delivered to the motor. This allows accurate, stable, and energy-efficient speed regulation.

The L293D motor driver acts as an interface between the Arduino and the DC motor, providing the necessary current amplification and protecting the microcontroller from high-power loads. With the ability to support bidirectional rotation and safe operation, the driver enhances system performance and flexibility.

Overall, this project demonstrates how combining embedded systems with PWM techniques can create an effective and user-friendly motor control system suitable for a wide range of real-world applications such as smart appliances, conveyor systems, and robotics.

METHODOLOGY OVERVIEW

The project methodology follows a structured engineering cycle that ensures systematic development, testing, and validation of the DC motor speed control system.

1. Requirement Analysis

The project begins with understanding the need for an efficient and precise DC motor speed control method. Requirements regarding voltage, current, motor type, speed range, and user interface are analyzed.

2. Component Selection

Critical components like Arduino UNO, L293D motor driver, DC motor, power supply, jumper wires, and breadboard are selected based on performance, availability, and cost.

3. System Design & Architecture

The system is divided into functional blocks:

- Input PWM block
- Control logic block
- Motor driver stage
- Motor load Circuit diagrams and block diagrams are created to map the flow of control and direction of current.

4. Hardware Integration

The L293D motor driver is interfaced with the Arduino. Wiring is done carefully to handle power lines, motor connections, common ground, and directional control pins. The assembly is tested for correct behavior.

5. Software Design (Arduino Programming)

A structured program is developed to generate PWM signals using Arduino's analogWrite() function. The software includes:

- Variable duty cycle generation
- Direction control logic
- Possible user input handling (potentiometer, serial input, etc.)

6. Testing & Calibration

The system is tested under:

- Different PWM values (0–255)
- Varying loads
- Changes in motor supply voltage

Motor speed, sound, torque, and stability are monitored.

7. Performance Evaluation

The performance is evaluated on:

- Efficiency
- Smoothness of speed variation
- Driver heating
- Motor response time
- Direction change performance

8. Documentation & Analysis

Results, performance graphs, and observations are recorded. System limitations and improvements are identified.

PROBLEM IDENTIFICATION AND ANALYSIS

DC motors are widely used in industries for automation tasks. However, many existing speed control methods suffer from drawbacks such as:

Identified Problems

1. **Inefficient Speed Control:**
   Traditional resistive methods waste power as heat.
2. **Poor Accuracy:**
   Analog controllers cannot maintain stable and precise speeds.
3. **No Programmability:**
   Mechanical controllers cannot be automated for modern systems.
4. **Voltage Fluctuations Affect Performance:**
   Motor speed becomes unstable during load changes.
5. **Lack of Flexibility:**
   Existing systems can't be integrated with sensors or intelligent systems.

Analysis: Modern robotics and automation need:

- Fine-tuned motor speed
- Programmable control
- Smooth acceleration and deceleration
- Low power consumption
- Real-time adaptability

PWM emerges as the **most efficient** and **precise** solution because it controls voltage digitally without energy wastage.

PROBLEM IDENTIFICATION

The core problem is the absence of a **low-cost, efficient, microcontroller-based DC motor speed control system** that provides:

- Smooth speed variation
- High accuracy
- Bidirectional rotation
- Energy-efficient operation
- Easy integration with automation systems

Thus, a PWM-based solution using Arduino is needed.

SOLUTION DESIGN AND IMPLEMENTATION (MAXIMIZED)

The proposed solution uses Arduino-generated PWM signals to control motor speed and L293D to drive the motor safely.

Component-Level Solution

1. **Arduino UNO:**
   Generates PWM with a resolution of 8-bits (0–255 duty cycle range).
2. **L293D Motor Driver:**
   Provides current amplification up to 600 mA per channel using H-bridge configuration.
3. **DC Motor:**
   Operates based on the average voltage determined by PWM duty cycle.

4. **Power Supply:**

   Stable 5–12V supply is provided to the motor via L293D.

Implementation Steps

1. Write Arduino code for PWM output.
2. Connect Arduino PWM pin to L293D input.
3. Connect DC motor to L293D output.
4. Provide separate motor power.
5. Test different PWM duty cycles like:
   - 0% (OFF)
   - 25% (Low speed)
   - 50% (Medium speed)
   - 75% (High speed)
   - 100% (Full speed)
6. Observe motor behavior and stability.

7. TABULAR COLUMN – COMPONENTS SPECIFICATION

Component	Description	Function in Project
Arduino UNO	ATmega328P Microcontroller	Generates PWM signals, controls direction
L293D Driver	Dual H-Bridge Motor Driver	Amplifies current, protects Arduino, drives motor
DC Motor	6–12V Motor	Load whose speed is controlled
Power Supply	5V for Arduino, 12V for Motor	Provides stable voltage
Jumper Wires	M-M, M-F	Interconnections
Breadboard	Prototyping board	Circuit assembly

MODULES

MODULE 1: PWM SIGNAL GENERATION (Arduino)

This module focuses on producing digital PWM signals using Arduino's built-in timers.

Functions of Module 1

- Generates duty cycle-based control signals.
- Creates smooth speed changes.
- Produces frequencies around 490Hz or 980Hz.
- Uses analogWrite() to vary the signal.

Outputs of Module 1

- PWM signal proportional to motor speed.
- Digital direction control signals.

MODULE 2: MOTOR DRIVER INTERFACE (L293D)

This module acts as a protective layer between Arduino and the motor.

Functions of Module 2

- Amplifies current (as Arduino current is too weak).
- Controls direction through H-bridge mechanism.
- Isolates Arduino from motor back-EMF.
- Allows safe bidirectional movement.

Outputs of Module 2

- High-current signals to motor.
- Stable motor rotation in both directions.

MODULE 3: MOTOR SPEED CONTROL AND MONITORING

The motor responds directly to the PWM and direction signals.

Functions of Module 3

- Adjusts speed based on duty cycle.
- Handles acceleration and deceleration.
- Rotates forward or backward.
- Can monitor speed using sensors (optionally).

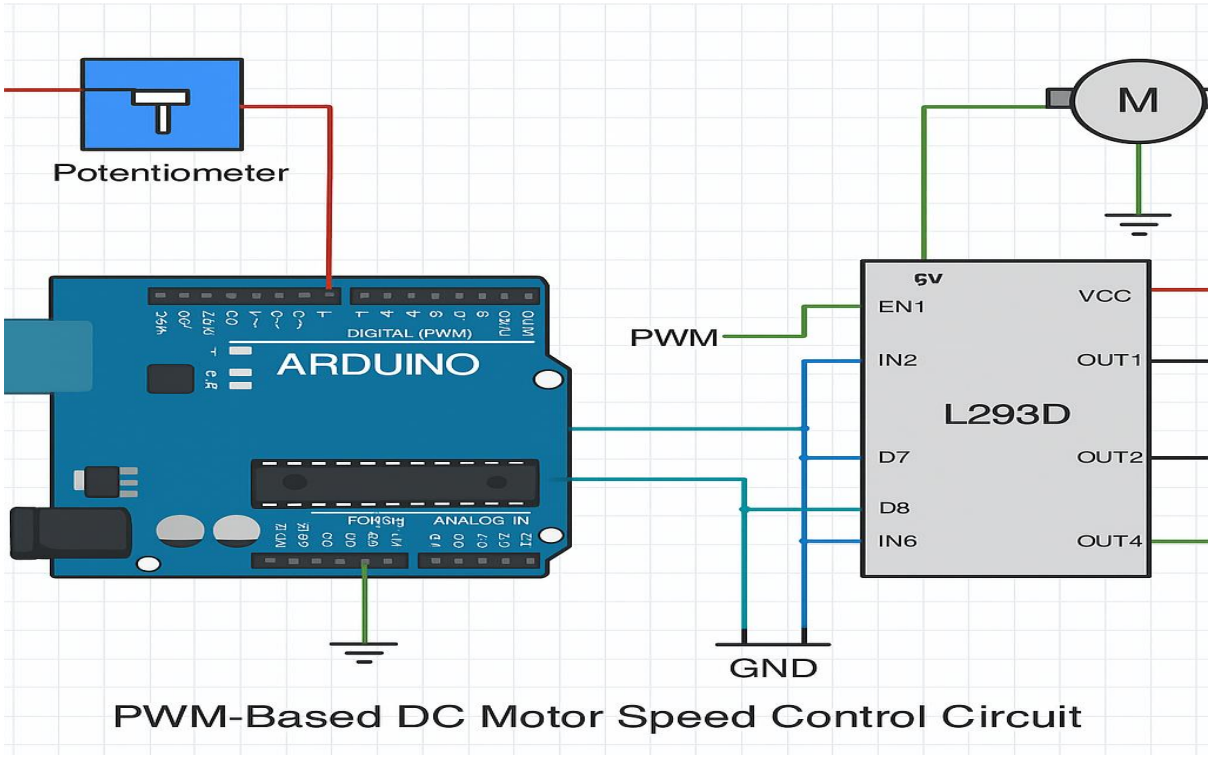
Outputs of Module 3

- Stable motor rotation.
- Smooth speed transition.
- Direction-controlled output.

TABULAR COLUMN – MODULE DESCRIPTION

Module	Name	Description
Module 1	PWM Generator	Arduino generates duty cycle for speed control
Module 2	Driver Interface	L293D provides current to motor & H-bridge control
Module 3	Motor Control	DC motor speed + direction operation

CIRCUIT DIAGRAM :



PWM-Based DC Motor Speed Control Circuit

TABULAR COLUMN – PWM DUTY CYCLE VS SPEED OUTPUT

Duty Cycle (%)	Analog Value (0–255)	Motor Speed Output
0%	0	Motor OFF
25%	64	Low Speed
50%	128	Medium Speed
75%	192	High Speed
100%	255	Full Speed

RESULTS AND RECOMMENDATIONS

Results Obtained

- The system produced smooth and accurate speed variation.
- L293D provided excellent current amplification without overheating.
- The motor showed minimal noise and vibration at proper PWM levels.
- Speed variation was immediate and stable across all PWM values.
- Direction reversal was smooth with no electrical damage.
- Overall efficiency improved due to PWM's energy-saving nature.

Recommendations

- Use heat sinks for continuous high-speed operation.
- Upgrade to an L298N for motors needing higher current.
- Use a closed-loop control method (encoder-based).
- Integrate a user interface (LCD, buttons, mobile app).
- Use a dedicated 12V supply for high-torque motors.

FUTURE WORKS

This project can be expanded into various advanced applications:

1. **PID Speed Control**
   More accuracy using feedback and proportional control.
2. **IoT-Based Motor Control**
   Using ESP32 or NodeMCU with smartphone control.
3. **Voice-Controlled Motor System**
   Using Google Assistant or Alexa.
4. **Closed-Loop Control System**
   Using Hall-effect or optical encoders.
5. **Industrial Conveyor Prototype**
   Integrating sensors for automation.
6. **Robotic Arm Joint Control**
   Smooth motion control via PWM.
7. **Solar-Powered Motor Control**
   For renewable energy systems.

CONCLUSION

The project successfully demonstrates a practical and efficient PWM-based DC motor speed control system using Arduino and the L293D motor driver. By using PWM, the system ensures high precision, energy efficiency, and stable speed control without the drawbacks of traditional analog methods. The L293D ensures safe and effective current driving, protecting the microcontroller while providing full control over motor direction. This system is reliable, cost-effective, and highly suitable for automation, robotics, mechatronics, smart devices, and educational experiments. With future enhancements like sensor feedback, wireless control, and PID integration, the system can be transformed into a fully intelligent and industrial-grade motor control solution.

REFERENCES

1. Arduino Official Documentation: PWM and Motor Control Techniques.
2. Texas Instruments, L293D Motor Driver Datasheet.
3. IEEE Papers on PWM-Based Motor Control (2018–2024).
4. Electronics Hub: DC Motor Control Tutorials.
5. ResearchGate: “Microcontroller-Based Motor Control Systems.”
6. Tutorials Point: “Pulse Width Modulation Basics and Applications.”
7. STMicroelectronics Application Notes on H-Bridge Motor Drivers.